

Fairness case study: COMPAS

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Case Study: COMPAS

- COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) is an AI application to determine the risk of prison in-mates re-offending (Recidivate) in the USA.
- It assigns a score from 1 to 10:
 - Low scores (1-4) means good chance of pre-trial release or parole;
 - High scores (5-10) means poor chance of pre-trial release or parole.
- So COMPAS has a real impact on prisoner's lives.
- COMPAS was trained with 100 factors including age, sex and criminal history, but not race.



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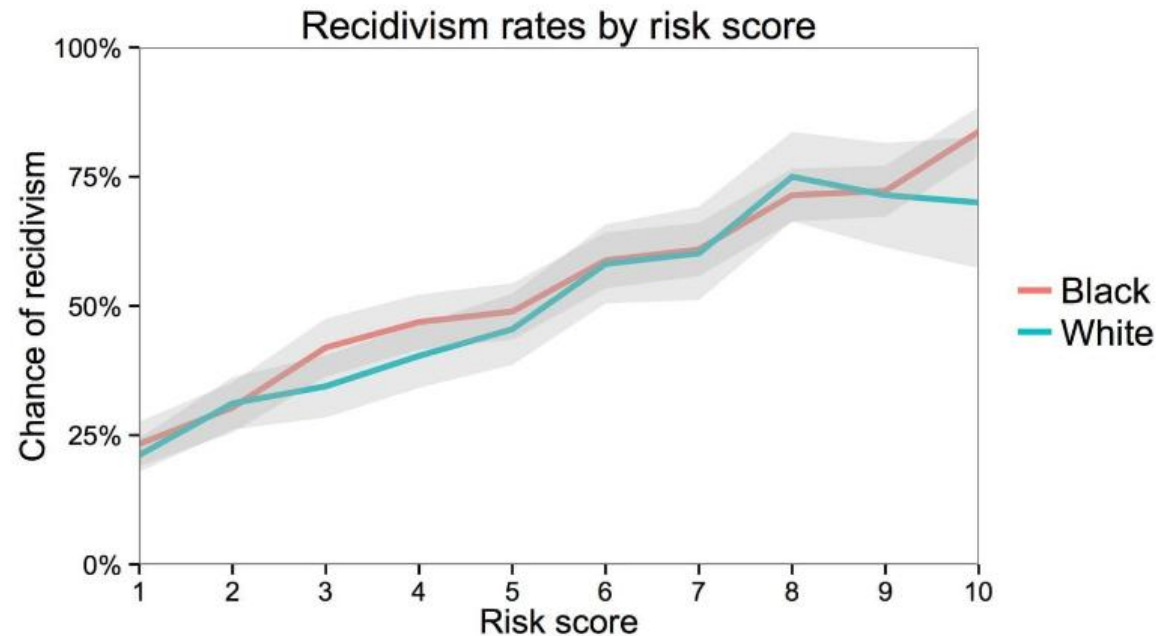
- In May 2016, news organization ProPublica claimed that COMPAS is biased against black defendants, since amongst those prisoners who subsequently did not re-offend, black people were much more likely to have been scored as high risk.
 - For non-reoffending black: 44.8% were given High score
 - For non-reoffending white: 23.5% were given High score
- This is not fair:
many more black non-reoffenders were given High score than whites.
- This is often cited as a case of AI Bias: e.g.
<https://www.prolific.com/resources/shocking-ai-bias>



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However...

- Northpointe, the company that created COMPAS responded by claiming their AI system is fair since if you take any score, the chance of reoffending is about the same regardless of race.



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- So who was right?
- How is it possible for both ProPublica to claim it was **unfair**, and Northpointe to claim it is **fair**?

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- For fairness analysis, the score is the response R (let's just say Low score: $R = 0$, High score: $R = 1$)
and Recidivated is the outcome ($Y = 1$ means Recidivated)
- The problem here is that ProPublica is using the fairness with **Separation** criterion (score conditioned on outcome), whilst Northpointe is using the **Sufficiency** criterion (outcome conditioned on score).
- It really depends which fairness measure you think is important.
- Wisconsin Supreme Court ruled in favour of Northpointe and COMPAS still runs.

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- Here is the data so you can do the analysis yourself:

Numbers		Score	
	<i>Outcome</i>	Low	High
Black	Survived	990	805
	Recidivated	532	1369
White	Survived	1139	349
	Recidivated	461	505

Numbers show number of prison in-mates within each category.

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Exercise

- Based on the table of data compute conditional probabilities for fairness metrics, as follows:
 - (1) Compute ProPublica results.
 - (2) Compute Northpointe results.
 - (3) Assess the data using the Independence criterion.
 - (4) Does the Independence criterion demonstrate fairness or unfairness?

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Some References

- J. Angwin, J. Larson, S. Mattu, L. Kirchner, Machine bias, in: Ethics of data and analytics, Auerbach Publications, 2022, pp. 254–264
<https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing>
- A. Rodà, The COMPAS case: an educational journey for explaining fairness in AI-based applications, AIMMES 2025 Workshop on AI bias: Measurements, Mitigation, Explanation Strategies
<https://ceur-ws.org/Vol-3961/paper4.pdf>